

Capability-Aware Compilation for Sectorized Observable Frameworks

Typed Admission and Cross-Species Registry Evidence

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This paper is Paper X of the RIME program. It develops capability-aware compilation and Registry evidence over the static and dynamic interfaces of Papers VIII and IX, while keeping compiler soundness distinct from adapter adequacy and automatic Registry ingestion.

Abstract

Problem. Paper VIII defines the static Sectorized Observable Framework (SOF), and Paper IX develops its dynamic architecture. This paper asks which conclusions a common interface may emit after a source adapter has declared its admission kind, capabilities, carriers, policies, and evidence. The challenge is to preserve these distinctions while compiling heterogeneous records into capability-sound reports.

Approach. We formulate a capability-aware compilation interface:

Capability Manifest $\longrightarrow$ Typed SOF IR

$\longrightarrow$ Profile-Gated Compilation $\longrightarrow$ Capability-Sound Compiler Output.

Strict admission factors through Paper VIII; dynamic fields are inherited from Paper IX. The Manifest declares availability; the IR records typed objects, findings, evidence, and derivations; and the Profile specifies Boolean module requirements and forbidden promotions.

Results. The Capability-Sound Compiler v1 Emission Theorem states that every affirmative claim item emitted by the algorithm has an eligible typed IR source and a valid derivation. Its claim-local carrier, convention, policy, evidence, and promotion checks must pass. Unavailable states, unresolved derivations, and forbidden promotions cannot produce affirmative conclusions. Strict SOF and diagnostic-analogue admission remain disjoint, and missing capability is not interpreted as zero. Registry v2.0 separately provides a source-addressed evidence interface with sparse capabilities and a reproducible census certificate.

Boundary. Compiler soundness does not establish scientific adequacy of an adapter, identify different carriers, or prove common cross-species dynamics. Compiler Output v1.0 is not yet a serialized SOFRS report. Registry v2.0 and Compiler v1.0 remain parallel compatible interfaces until a versioned Registry-to-IR adapter is supplied.

Notation Table

Symbol	Meaning
$\mathcal{F} = (V, Q, Y; X, \mathcal{H})$	typed Sectorized Observable Framework, with optional Lie/Hall filtration $\mathcal{H} = \{\mathcal{H}_d\}_{d \geq 0}$
$\mathcal{R}$	SOF Registry
$\text{Cap}(\mathcal{E})$	capability declaration of a Registry entry
strict SOF	finite complex (V, Q, Y) realization satisfying structural admission
diagnostic analogue	source descriptor and analogue mapping without a strict SOF theorem instance
$R_1[Y], \text{Route}_d[Y], W_d[Y]$	operator, route, and word channels
$D_{\text{route}}[Y], D_{\text{word}}[Y]$	typed route and word depths
$R_1^{\text{Lie}}, R_2^{\text{Lie}}, D_{\text{Lie}}$	independently registered Lie/Hall channels
K_0, K_1, K_2	raw norm proxies for direct, commutator, and nested-depth observables

1 Introduction

Papers VIII and IX establish the static SOF object language and its dynamic trajectory layer, respectively [9, 8]. This paper addresses two interface questions:

1. How can different source systems enter a single typed reporting interface without forcing them into the same carrier, diagnostic, or claim status?
2. Which conclusions may be emitted once admission, capabilities, policies, and evidence have been declared?

This compilation interface is not a new structural decomposition theorem.

Classical Wedderburn–Artin theory decomposes semisimple representations into structural blocks [11, 16, 13]. This paper starts above that layer and asks how sectorized observables, typed channels, dynamics, and diagnostics are organized once a structural or native representation has been chosen.

The guiding principle is:

different sources, explicit typed admission, capability-gated compilation, and no unsupported promotion.

This paper treats capability-sound compiler emission as the theorem-level object. The Registry is compatible evidence architecture: it records which sources are admitted, which capabilities and carriers are declared, and which findings are validated. Registry v2.0 and Compiler v1.0 are parallel interfaces here. No versioned Registry-to-IR adapter is specified, so a Registry row does not by itself enter a Report Profile compilation.

Registry admission is capability-sparse. A source enters the strict branch only after satisfying the realization criteria. Sources that expose useful structural descriptors without a complete finite-dimensional complex realization (V, Q, Y) may enter only as diagnostic analogues.

2 Related Work and Novelty Boundary

Program lineage. This paper builds on the SOF object language of Paper VIII [9] and the observable-dynamics language of Paper IX [8]. Registry entries used here reuse the fixed spectral geometry of Paper IV [5], the accessibility calculus of Paper V [4], the normality-gated point

registrations and linearized moving-field interface of Paper VI [7], and the projected-composition incidence geometry, rank-protection results, and promotion limits of Paper VII [6].

External background. The representation-theoretic structural layer is the standard finite-dimensional semisimple/Wedderburn–Artin background [11, 16, 13]; SOF adds an observable layer above that decomposition. The rate-separation precedent is Xu–Vardi–Safran’s ridge-regression grokking model [18]. Dynamic precedents for plateaus and wall crossing belong to the Paper IX deformation layer and are cited there [8]; this paper uses only the resulting typed findings and policies.

Compiler and evidence interfaces. Type and effect systems motivate the separation of object identity, available operations, and checked effects [3, 14]. Schema evolution work clarifies why a change of required fields or meanings needs an explicit version transition rather than silent reinterpretation [1]. Scientific provenance provides the source, transformation, and digest discipline used by the artifact graph [17, 2]. Model cards and datasheets provide precedents for evidence-aware reporting and explicit scope boundaries [15, 12]. This work does not reproduce those systems; it specializes their interface concerns to carrier-qualified mathematical claims and proof- or certificate-sensitive promotion.

Novelty boundary. This paper does not redefine the static objects of Paper VIII or the dynamic fields of Paper IX. Its contribution is the capability-sound compilation layer and its Registry evidence architecture: validated declarations, typed findings, evidence-qualified derivations, and graceful degradation determine which claim items a compiler may emit.

3 Capability-Aware Observable Compilation

3.1 Pipeline Interface

The common structure behind the registered examples is not a shared native coordinate system. It is a shared pipeline:

$$\begin{aligned} \text{source system} &\longrightarrow \text{adapter and admission} \longrightarrow \text{Capability Manifest} \\ &\longrightarrow \text{Typed SOF IR} \longrightarrow \text{Report Profile} \longrightarrow \text{Compiler Output v1.0.} \end{aligned}$$

The common object is a compilation interface, not a shared result tuple. Strict records factor through the static SOF core; diagnostic analogues use a separate mapping and negative boundary. Both can use the evidence interface, but only validated capabilities can enable report modules.

The compiler pipeline is implemented by the contract fixtures listed in Appendix A. Although Registry v2.0 uses compatible typed fields, it is not an input fixture for that implementation: its `capability_manifest` and `typed_sof_ir` contract references remain intentionally null until a versioned Registry-to-IR adapter is defined. The Registry census certificate and compiler-contract certificate are therefore separate Computational Certificates.

The compiler output is a typed collection of affirmative claim items and degradation items. It is not yet a versioned SOFRS report artifact; serialization, presentation, and report-level epistemic metadata belong to Paper XII [10].

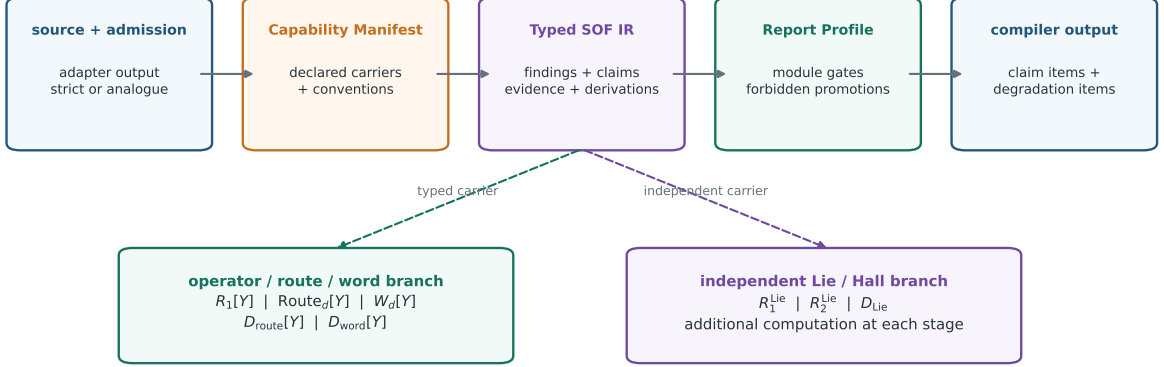
3.2 Construction 1 (Strict Admission Factorization)

Let a source $\mathcal{S}$ supply:

1. a finite-dimensional complex Hilbert space V ;

Capability-Sound Compiler v1 Emission

Profiles select validated IR claims; they do not create evidence or cross carriers.



NOT_DECLARED, unresolved derivations, and forbidden promotions emit no affirmative claim.

Figure 1. Capability-sound observable compilation. Source adapters produce an admission record, Capability Manifest, and Typed SOF IR. Compiler v1 emits only claims selected by the Report Profile and supported by declared capabilities, policies, and evidence.

2. a complete labelled orthogonal sectorization $\{Q_i\}$;
3. a finite observable extraction rule producing a labelled operator family Y , together with any separately declared Lie/Hall registration $(X, \mathcal{H})$, where $\mathcal{H} = \{\mathcal{H}_d\}_{d \geq 0}$ is the declared formal-expression filtration and depth convention.

Relative to the declared choices, these data determine the typed SOF core

$$\mathcal{F}_s = (V, Q, Y; X, \mathcal{H}),$$

and the static constructions supported by its declared enrichments. Here the entries after the semicolon are omitted when no Lie/Hall enrichment is declared. The definitions of the linear layers, closures, filtrations, and optional Lie/Hall carrier belong to Paper VIII [9]; they are not repeated here. This factorization does not itself declare a route, word, Lie/Hall, deformation, closure-certificate, or report capability.

3.3 Compiler v1 Semantics

The compiler is bound to **Capability Manifest v1.0**, **Typed SOF IR v1.0**, and **Report Profile v1.0**. Its output item type is the disjoint union

$$\text{CompilerItem}_{v1} = \text{ClaimItem}_{v1} \sqcup \text{DegradationItem}_{v1}.$$

Thus

$$\text{Compile}_{v1}(M, I, P) \in \text{List}(\text{CompilerItem}_{v1}),$$

with projections `AffirmativeClaims` and `Degradations`. Claim items retain the emitting module, source IR claim, carrier references, result state, evidence level, and derivation references. Degradation items retain the omitted module or claim and the gate that failed.

For a typed IR source package x and candidate claim c , write

$$\text{Eligible}_{M,P}(x, c)$$

when the following checks pass: admission, module capability, carrier, convention, claim-local policy, result state and evidence, and promotion permission. This predicate does not assert that c follows from x . Write

$$I \vdash_{\text{derive}} x \Rightarrow c$$

for the separate derivation relation generated by identity emission and valid v1.0 rule-registry derivations whose hypotheses and condition checks are satisfied. Open rules and unresolved derivations do not belong to this relation. Finally, write $I \models_{\text{scope}} c$ when the typed IR and a sound registered derivation support c with its declared carrier, scope, hypotheses, and evidence level. For a Computational Observation, this denotes bounded evidential support rather than theorem status.

The executable Compile_{v1} validates the contract triple, traverses profile modules in declared order, emits a degradation item for a blocked module or ineligible selected claim, and emits a claim item only after both $\text{Eligible}_{M,P}(x, c)$ and $I \vdash_{\text{derive}} x \Rightarrow c$ succeed.

Theorem 3.1 (Capability-Sound Compiler v1 Emission).

Let M be a schema-valid Capability Manifest v1.0, let I be a Typed SOF IR v1.0 whose references and result-state/evidence pairs validate against M , and let P be a Report Profile v1.0 applicable to the admission kind of M . Assume that every theorem-status rule used by $\vdash_{\text{derive}}$ is sound within its registered scope. If

$$c \in \text{AffirmativeClaims}(\text{Compile}_{v1}(M, I, P)),$$

then there is a typed IR source package x such that

$$\text{Eligible}_{M,P}(x, c) \quad \text{and} \quad I \vdash_{\text{derive}} x \Rightarrow c.$$

Consequently $I \models_{\text{scope}} c$. In particular, a `NOT_DECLARED` or `NOT_APPLICABLE` state, an unchecked or failed derivation condition, or a forbidden promotion cannot generate an affirmative claim item. A diagnostic analogue cannot instantiate a strict-SOF theorem claim.

Proof. Proceed by induction over the ordered module-emission rules of Compile_{v1} . If record-kind applicability or a module's Boolean capability, object, convention, or run-policy expression fails, the corresponding rule emits only a degradation item, so it contributes no element to AffirmativeClaims .

Consider an enabled module. The candidate-selection rule first filters by carrier kind, result state, and claim status. It then re-evaluates the module expressions using only the capability, object, convention, and run-policy references carried by the candidate and its typed finding dependencies. A globally present threshold or certificate therefore cannot satisfy an omitted claim-local reference. The evidence rule requires a proof reference for a Theorem, a passing certificate for a Computational Certificate, and a source artifact for a Computational Observation. The promotion rule rejects every forbidden promotion ID. These checks establish $\text{Eligible}_{M,P}(x, c)$.

For identity emission, $I \vdash_{\text{derive}} x \Rightarrow c$ holds directly. For a derived claim, the emission rule requires a non-open registered rule, a `VALID` derivation state, and satisfied proof- or certificate-backed conditions. The induction hypothesis applies to its source packages, and the assumed scoped soundness of the registered rule establishes the target claim. Therefore every claim-emission branch supplies both the eligibility witness and the derivation witness. No degradation branch emits an affirmative claim, which proves the statement.

For an exact finite depth claim $D_\kappa = d$, $\text{Eligible}_{M,P}(x, c)$ requires a first-hit certificate: a level- d witness together with verified non-hits at all levels $1, \dots, d-1$. A level- d witness without

that minimality audit supports only $D_\kappa \leq d$, or the finding that a hit was observed by level d . Non-attainment through a finite cutoff remains the distinct state `UNREACHED_AT_CUTOFF`. Likewise, trajectory non-crossing on a bounded observation interval is response censoring carried by its sampling or trajectory policy; it is not this filtration state.

The theorem applies only to the v1.0 semantics of the three named input contracts and the v1.0 emission algorithm. The schemas, rule registry, validator, compiler-output fixture, and boundary regressions in Appendix A (A1–A5) provide a Computational Certificate that the implementation realizes these emission rules; that certificate does not replace the proof. A future major contract version requires the algorithm and soundness theorem to be restated or reproved. $\square$

3.4 Claim Spine

This paper separates role and ownership from the four reader-facing claim statuses.

Item	Role / ownership	Claim status	Content
Strict Admission Factorization	inherited construction from Paper VIII	not an independent Paper X claim	finite complex (V, Q, Y) data determine a typed SOF core relative to declared choices
Capability-Sound Compiler v1 Emission	owned result	Theorem	the algorithm emits only claim items with eligible IR sources and valid derivations
Registry v2.0 census	owned finite evidence	Computational Certificate	snapshot counts are bound to the Registry content digest, schema, and validator

3.5 Registry Probes and Open Dynamic Directions

The cross-species evidence motivates the following rate-hierarchy target but does not place it in the compiler claim spine:

$$\tau(R_1^{\text{Lie}}) < \tau(R_2^{\text{Lie}}) < \tau(D_{\text{Lie}}^{(\leq d_{\max})}).$$

An exact-depth version with D_{Lie} is a stronger future target and requires a closure or saturation certificate. This is a Paper IX-compatible dynamic Research Program recorded as a Registry probe, not an owned compiler result.

The commonality developed here is therefore not common dynamics. It is a capability-sound compilation interface. Different species may possess completely different deformation geometries. What is shared is the typed passage from a domain source through strict or analogue admission and capability declarations to validated findings. Observable dynamics remain species-dependent.

There is also an explicitly programmatic direction: the **Observable Sufficiency Program**. Its question is whether an expanding class of registered species can be compared using aligned typed fields, including operator/word and Lie/Hall branches when independently registered. This is a research program for future alignment work, not a theorem established here or part of the compiler spine.

3.6 Inherited Sector Boundary

This paper uses the Paper VIII No-Sector No-Shadow boundary [9]: sector-indexed findings require declared projectors, while global observables may exist without them. A source lacking the required marked sectorization cannot validate as a strict-SOF Manifest. If that source is otherwise admitted under another record kind, sector-indexed modules degrade rather than emit claims. The compiler does not infer that every useful observable theory must be sectorized.

3.7 Capability-Aware SOF Registry

Definition 3.2 (SOF Registry).

A **SOF Registry** is a finite or open-ended collection

$$\mathcal{R} = \{\mathcal{E}_s\}_{s \in S}$$

of typed evidence entries. Each entry $\mathcal{E}_s$ first declares one of two admission kinds:

1. a **strict SOF realization**, with a finite complex space, a complete marked sectorization, and a labelled operative alphabet (V_s, Q_s, Y_s) ;
2. a **diagnostic analogue**, with source descriptors, an explicit analogue mapping, provenance, and a negative boundary excluding strict SOF theorem instantiation.

The entry then supplies a capability declaration $\text{Cap}(\mathcal{E}_s)$, typed objects and carriers, semantic conventions, run policies, channels, structured findings, claims, certificates, artifacts, and audited derivation edges. Dynamics are optional and must identify their comparison and trajectory data when required by the selected diagnostic.

The Registry is therefore a compatible evidence interface, not a fixed result tuple or an already compiled Manifest/IR pair. A row need not supply route, word, closure, Lie/Hall, deformation, wall, or response data unless those capabilities are declared. Conversely, declaring one carrier does not authorize a nearby one: a graph path cannot fill a routed product, a star closure cannot fill a positive-word depth field, and a proxy cannot fill a discrete Lie shadow.

Sector origin is retained as provenance rather than redefined here. Representation-, control-, mesh-, graph-, and Dirac-derived coordinates may instantiate the Paper VIII realization interface when the strict structural admission conditions are satisfied. Stochastic-barrier, activation-derived, and other source coordinates may instead supply diagnostic-analogue mappings. Sector origin is retained as provenance in both branches, but only strict entries instantiate a Paper VIII SOF. Evidence level records support for a finding; it does not determine the admission kind.

Every sector-pair channel also records its pair scope. In particular, a saturated-corner channel must be either off-diagonal or scalar-reduced on the diagonal. Indeed, because all three generated algebras are unital,

$$Q_i \in Q_i A_Y^+ Q_i \cap Q_i A_Y^* Q_i \cap Q_i A_{Q,Y}^* Q_i.$$

Raw diagonal saturated Boolean support is therefore identically true and is not an informative Registry coordinate. A nontrivial diagonal channel must remove $\mathbb{C}Q_i$, for example by the Hilbert–Schmidt orthogonal complement, and declare that reduction in its channel semantics.

Registry records are capability-sparse. A missing capability, a computed zero, a cutoff-unreached state, a validation failure, and mathematical nonexistence are distinct states. A compiler may omit an unsupported module or report it as unavailable, but it may not manufacture the missing field.

Thus the Registry is the evidence architecture developed here. The theorem-level object is sound compilation through the interface; the Registry records instances, controls, and boundary cases.

3.8 Diagnostic Vocabulary

The shared diagnostic vocabulary is:

Open SOF Registry

Different sources enter a common evidence architecture without sharing a deformation law.

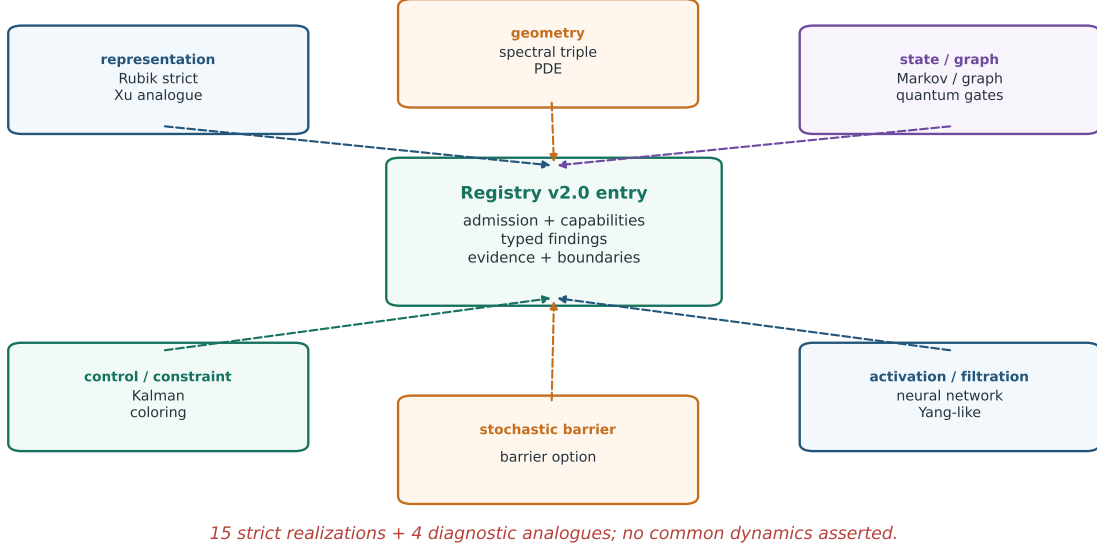


Figure 2. Registry wheel. Strict realizations and diagnostic analogues enter through different source maps, including representation, gate action, state partitions, control or mesh geometry, Dirac blocks, stopping regions, activation descriptors, and filtrations. The wheel records admission and evidence roles; it does not assert common dynamics.

Diagnostic	Meaning
$\tau(O)$	characteristic time scale
$\text{wall}_{\text{ref}}(O; \mathcal{C}, P_W)$	imported typed-wall reference under declared context and policy
$\text{Repair}_{\text{filt}}^{\kappa, (\leq d_{\text{max}})}$	static finite-filtration recovery in carrier κ under a declared cutoff and pair scope
$\text{Repair}_{\text{dyn}}^{\kappa}$	time- or parameter-indexed shadow transition in carrier κ under a declared chart and trajectory
$\text{Repair}_{\text{an}}^{\kappa}$	carrier-qualified diagnostic-analogue repair descriptor with no strict SOF theorem instantiation
$\text{plateau}_{\text{diag}}(O; \gamma, P_{\tau})$	trajectory-relative stable-regime diagnostic

These repair kinds are not interchangeable. Every repair finding records its carrier, source and target layers, temporal scope, predicate, cutoff, pair scope, count denominator, and saturation status. Not every diagnostic is defined for every species; absence is recorded rather than hidden. The wall and plateau entries import Paper IX admissibility and Paper XI morphology rather than defining generic wall or plateau objects here.

3.9 Registry v2.0 Interface

Registry v2.0 uses the following entry-assembly order:

source $\longrightarrow$ admission and capabilities
 $\longrightarrow$ typed objects, carriers, and policies
 $\longrightarrow$ findings and evidence.

This is the observable layer added above classical structural decomposition. Wedderburn–Artin theory explains how semisimple representations decompose into structural blocks [11, 16, 13]. The SOF Registry begins after this decomposition stage: it records how sectorized

Capability-Aware Registry v2.0 Entry

Admission and capabilities determine which findings may be registered.

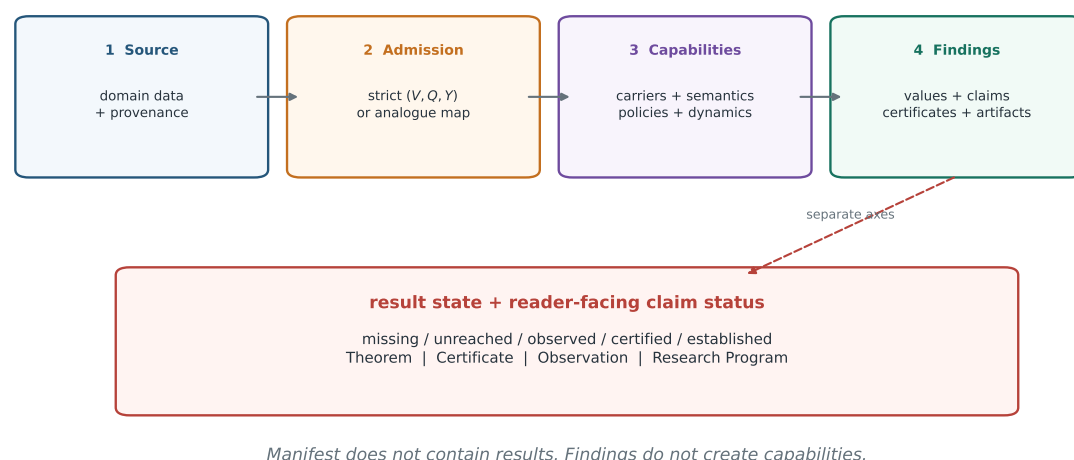


Figure 3. Capability-aware Registry entry. A source is admitted either as a strict SOF realization or as a diagnostic analogue. Declared capabilities gate typed carriers and reportable findings; evidence and negative boundaries prevent unsupported promotion.

observables, typed transport channels, deformation geometries, and diagnostics are built on the represented or finite system. The sectorization-origin field records where the coarse coordinate system comes from; it is Registry metadata, not an additional SOF axiom.

The complete registry is intentionally larger than the main argument. To keep the discussion centered on the pipeline rather than on a catalogue, the main text uses a compressed capability view. Appendix A (A6–A8) identifies the immutable v1.0 snapshot, the typed Registry v2.0 snapshot, and its schema and validator.

3.10 Registry Census Certificate

The following counts have status **Computational Certificate**, not Theorem. The snapshot stores a `census_certificate` containing its snapshot ID, a SHA-256 digest of the canonical Registry payload excluding the certificate itself, schema version and digest, validator ID and version, validation status, query version, and recomputable summary. The validator rejects a stale digest, count mismatch, schema mismatch, or validator-version mismatch. Appendix A (A9–A11) identifies the generated evidence record, its producer and validator, and the migration regression.

Registry v2.0 coordinate		Count	Meaning
rows	19		versioned evidence entries
strict SOF realizations	15		complete finite complex (V, Q, Y) admission
diagnostic analogues	4		mapped descriptors without strict theorem instantiation
route / word / Lie–Hall carriers	1/4/5		optional finite-filtration capabilities
proxy diagnostics / typed deformation charts	7/2		proxy capability is broader than deformation admissibility
declared closure capabilities	0		no closure conclusion is inferred from finite-depth data
findings by evidence level	1 Theorem, 13 Computational Certificates, 14 Computational Observations		28 structured findings in total

This compressed table is the reader view of that certificate. It is evidence that the interface admits multiple species without forcing them into the same carrier or evidence level; it is not the compiler-soundness theorem.

The finite spectral-triple entry plays a distinct logical role. It is conceptually different from the representation-based examples: its sectorization is induced by a block-diagonal Dirac operator rather than by irreducible representation theory or Wedderburn–Artin block decomposition. Nevertheless, the same observable pipeline yields bridge-level accessibility diagnostics. This suggests that the SOF pipeline depends primarily on the existence of a compatible sectorization, rather than on the algebraic origin of that sectorization.

4 Registry Evidence: Cross-Species Capability Probes

4.1 Markov and Graph: Static Positive-Word Registrations

Two foundational strict rows use coordinate sectors and a single declared operative letter. The Markov row uses a three-state column-stochastic lazy directed cycle with alphabet $\{P\}$; the graph row uses the path graph P_6 with vertex sectors and adjacency alphabet $\{A\}$. Both declare operator and positive-word carriers, but no Lie/Hall carrier.

The exact finite audit gives

Registration	$R_1[Y]$	$W_2[Y]$	$\max D_{\text{word}}[Y]$
Markov lazy cycle	3/6	6/6	2
path graph P_6	10/30	8/30	5

The denominators count ordered off-diagonal sector pairs. Every such pair is eventually reached in both registrations. Because each example has one entrywise nonnegative operative matrix, a positive matrix entry in a power cannot be removed by route-sum cancellation. Word support therefore agrees with support-graph walks for these two declared matrices, and finite shortest-path comparison certifies the displayed exact depths. This is a realization-specific certificate, not a general promotion from graph paths to operator words.

The graph row also carries a separate rewiring K_0/K_1 proxy observation, discussed below. That dynamic proxy neither defines nor alters the static $R_1[Y]$, $W_2[Y]$, or $D_{\text{word}}[Y]$ certificates.

4.2 Xu–Vardi–Safran Grokking: Rate-Separation Probe

Xu–Vardi–Safran’s ridge-regression analysis [18] separates a parameter vector into row-space and null-space components:

$$\theta = \theta_{\parallel} + \theta_{\perp}.$$

The row-space component is driven by the empirical loss and evolves quickly; the null-space component is controlled by weight decay and evolves slowly. This is not an SOF theorem. Registry v2.0 admits it as an external diagnostic analogue: the source theorem remains external, while the Registry records only the mapped rate descriptor and its negative strict-SOF boundary.

The conceptual bridge to SOF is the transition from parameter-space rate separation to an observable-space rate question. The Registry evidence audit reports:

Domain	Fast channel	Slow channel	Ratio	Evidence type
Ridge regression	$\theta_{\parallel}$	$\theta_{\perp}$	$749.6\times$	locally derived contraction-deficit ratio from an external-theorem diagnostic analogue
Exact three-sector trajectory	selected direct block norm	selected simple-commutator block norm	$\eta^{-1/2}$ at common threshold $0 < \eta < 1$	Inherited Paper IX Theorem relative to the declared trajectory and policy
NN training diagnostic analogue	K_0	K_2	ordered half-response $60 < 80 < 120$	Computational Observation

This evidence is structural; it does not assert numerical equality of the ratios. Different domains produce different magnitudes, and the three-sector ratio varies with the declared threshold. Response times are compared only relative to a stated trajectory parameterization, observable normalization, norm, and policy.

The ridge value is not source-reported and is not a half-response-time ratio. It is locally derived by the registered seeded diagnostic as

$$\frac{\eta(\lambda_{\min}^+ + \lambda)}{\eta\lambda} = 749.6,$$

using $m = 400$, $n = 50$, $\eta = 0.2$, $\lambda = 10^{-4}$, and seed 42. Registry v2.0 records the source locator, extraction formula, parameter values, response convention, and `locally_derived` status. The external paper supplies the theoretical precedent; it does not directly report this Registry number.

4.3 Yang-Like State Mixing: Proxy Boundary

The active Yang-like row studies a declared state-mixing path

$$\rho(\varepsilon) = (1 - \varepsilon)\rho_0 + \varepsilon\sigma.$$

and evaluates two continuous block-norm proxies. Their half-response audit is degenerate rather than ordered. Registry v2.0 therefore admits this row as a diagnostic analogue and proxy boundary, not as a strict deformation chart or typed wall. The earlier plateau/oscillation script is retired historical provenance and supplies no active Registry v2.0 finding.

The registry lesson is:

common evidence interface, different deformation geometry, different proxy behavior.

Rate-Separation Evidence Has Different Claim Status

Shared qualitative ordering does not identify carriers, mechanisms, or numerical scales.

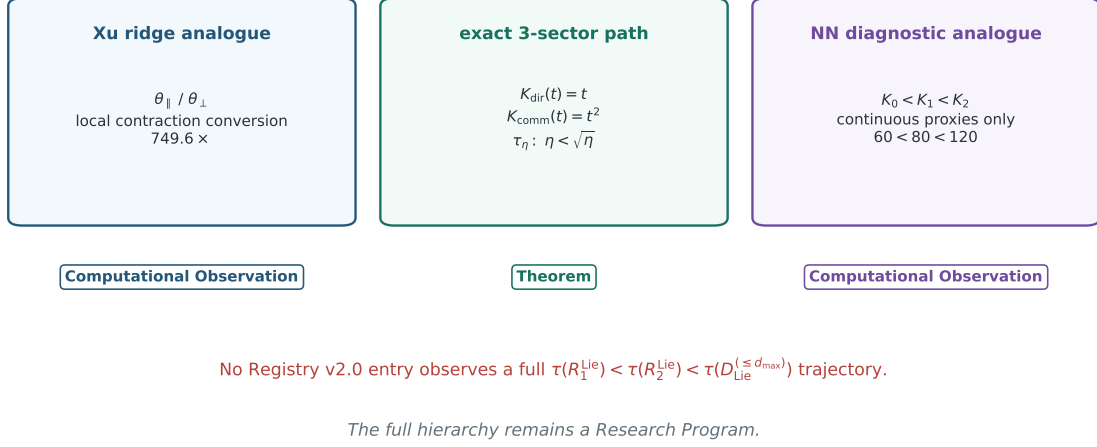


Figure 4. Rate-hierarchy evidence. Xu ridge dynamics, NN proxy diagnostics, and an exact three-sector threshold construction provide three evidence levels for hierarchical visibility. Their claim status and parameter semantics differ.

4.4 Quantum Clifford/CNOT: Non-Rubik Accessibility Probe

Quantum gate systems provide a non-Rubik accessibility probe. In small computational-basis sectorizations, this certificate uses the unique Registry carrier `quantum.gates.principal-log-skew.hall-v1`. Its two-qubit generator list is $\{H, S, \text{CNOT}\}$, with H and S acting on the first qubit. For each embedded gate U , the registration applies the SciPy principal matrix logarithm and then the skew-Hermitian projection $X = (\log U - (\log U)^*)/2$, with no additional normalization. The source code declares direct skew extraction $(U - U^*)/2$ only as an exception fallback. The diagnostic observations are:

1. entangling generators such as CNOT reduce frozen pairs and open additional transport channels;
2. the T gate does not enrich the tested low-order R_1^{Lie}/R_2^{Lie} support beyond the corresponding Clifford+CNOT gate set;
3. the same typed Lie/Hall audit can be applied outside Rubik.

This entry uses a **static filtration repair** count, not a dynamic transition. For ordered off-diagonal sector pairs define

$$\text{Repair}_{\text{filt}}^{\text{Lie},(\leq 4)}(i, j) = \mathbf{1} \left[R_1^{\text{Lie}}(i, j) = 0, \quad D_{\text{Lie}}^{(\leq 4)}(i, j) \neq \text{unreached} \right].$$

The registered predicate imposes no $R_2^{\text{Lie}} = 0$ condition. The Hall filtration places generators at depth 0, simple commutators at depth 1, and generator brackets with the preceding layer at later depths, modulo the cumulative real-linear span. The registered cutoff field 4 denotes the four tested levels with engine indices 0, 1, 2, 3; the absolute zero tolerance is 10^{-6} . The two-qubit computational basis gives 12 ordered off-diagonal pairs, with

$$N_{\text{repair}, \text{filt}}^{\text{Lie},(\leq 4)}(\text{Pauli}) = 0/12, \quad N_{\text{repair}, \text{filt}}^{\text{Lie},(\leq 4)}(\text{Clifford+CNOT}) = 6/12.$$

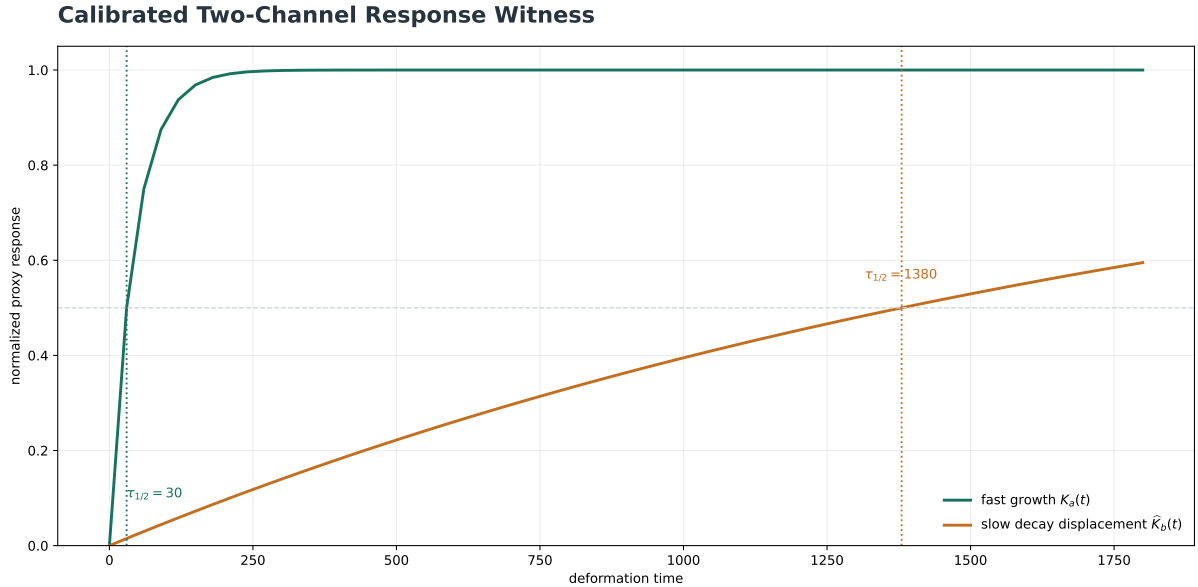
Thus, the value 6 is a static pair count, not a depth. The registration is truncated-only and asserts neither saturation nor a time-dependent repair event. Artifact A9b records the carrier ID, generator extraction, normalization, Hall convention, cutoff indexing, tolerance, and pair scope; no alternative gate-generator registration belongs to this certificate.

This is not a claim that quantum systems obey Rubik wall theory. It is a registry claim: quantum systems can expose a typed Lie/Hall channel under a different species geometry.

5 Imported Dynamic Evidence

This paper does not re-prove response-time results. From Paper IX, it imports the calibrated exponential theorem and its three-sector Computational Certificate with half-response times $30 < 1380$, together with the direct/commutator threshold construction and NN proxy observations [8]. The Registry preserves their distinct theorem, certificate, and observation statuses. Here, import means registration in the Paper X evidence interface; it does not make the Paper IX artifacts Compiler v1 input fixtures or implement a Registry-to-IR adapter.

The compiler-level boundary is that none of these findings may be promoted to a Boolean support time, dynamic-shadow repair time, or Lie depth without the missing proxy-to-shadow conditions identified in Paper IX. The static Clifford+CNOT repair count remains a separate Registry certificate and cannot fill that dynamic field.



One normalized-displacement policy; theorem for ordering, certificate for block realization.

Figure 5. Imported calibrated response fixture. Paper IX owns the response theorem and block-realization certificate; this paper preserves their typed status in the Registry.

5.1 Evidence Stratification

Evidence layer	Status	Interpretation
Calibrated exponential response model	Inherited Paper IX Theorem	exact result under one normalized-displacement policy
Three-sector block-norm realization	Inherited Paper IX Computational Certificate: $30 < 1380$	registered typed dynamic finding, not yet compiled through a Registry-to-IR adapter and not a cross-species causal law
Exact three-sector threshold construction	Inherited Paper IX Theorem: $\tau_\eta(K_{\text{dir}}) = \eta$ and $\tau_\eta(K_{\text{comm}}) = \sqrt{\eta}$ for $0 < \eta < 1$	first- versus second-order norm scaling under the declared trajectory and policy; not an intrinsic rate invariant, Boolean support, or Lie depth
NN training diagnostic analogue	Computational Observation: $60 < 80 < 120$	endpoint-normalized sampled proxy response hierarchy; no strict SOF theorem instance
Markov and graph strict SOFs	Computational Certificates	direct and length-two positive-word support plus exact first word depth for two declared entrywise nonnegative matrices; no Lie/Hall carrier
Finite spectral-triple strict SOF	Computational Certificate plus Computational Observation	central distance obstruction and two ordered bridge instances remain separate
Quantum strict SOF	Computational Certificate: static filtration-repair count 6/12 within cutoff	ordered-pair count, not depth 6 or a response trajectory
Paper VI Rubik registration	Computational Observation: pointwise R_1^{Lie} only	no R_2^{Lie} , D_{Lie} , or moving-field positive instance is imported
Xu ridge diagnostic analogue	Computational Observation: locally derived contraction-deficit ratio $749.6\times$	external theorem is precedent; the registered number is not source-reported or a response-time ratio
Graph-rewiring / Yang-like proxy boundaries	Computational Observations	edge rewiring and state mixing are degenerate or unordered under their declared policies
Proxy-to-shadow bridge	Research Program	Observable Proxy Shadow Principle not yet proved
Dynamic truncated- D_{Lie} repair trajectory	Research Program	Clifford+CNOT gives a static filtration-repair count 6/12, but no dynamic-shadow repair or structured $\tau(D_{\text{Lie}}^{(\leq d_{\text{max}})})$ audit

The inherited response proposition is an exact calibrated model, not a blanket causal theorem. Its realization certificate uses one strict proxy-capable deformation chart. The NN, Xu, and Yang-like rows are diagnostic analogues; the quantum row is a static Lie/Hall audit; and the graph row contains separate static positive-word certificates and a rewiring proxy boundary. These roles are not interchangeable. A response comparison is meaningful only after its parameterization, normalization, norm, and response policy are fixed.

6 Claim-Status Boundary

This paper claims:

1. Capability-Sound Compiler v1 Emission is the central theorem-level object;
2. strict admission factors through the Paper VIII SOF core rather than constituting a new realization theorem;

3. the SOF Registry is capability-aware evidence architecture with separate strict-SOF and diagnostic-analogue admission;
4. the compilation interface inherits the Paper VIII No-Sector No-Shadow boundary;
5. registered rows may be compared only through aligned fields with compatible carriers, conventions, policies, and evidence levels;
6. the Registry Census Certificate records 19 rows, including 15 strict realizations and 4 diagnostic analogues, with a recomputable digest and schema/validator binding.

This paper does not claim:

1. every mathematical system is naturally an SOF;
2. every SOF has a canonical nontrivial deformation;
3. a diagnostic analogue is a strict SOF theorem instance;
4. a missing capability is equivalent to zero, cutoff-unreached, failed validation, or mathematical nonexistence;
5. every row supplies route, word, closure, Lie/Hall, deformation, wall, and response-time fields;
6. all SOFs obey $\tau(R_1^{\text{Lie}}) < \tau(R_2^{\text{Lie}}) < \tau(D_{\text{Lie}}^{(\leq d_{\text{max}})})$;
7. Registry v2.0 contains a dynamic-shadow repair-rate trajectory or that its static filtration-repair count is a depth value;
8. the continuous proxy family $K_0/K_1/K_2$ already determines the discrete Lie shadows R_1^{Lie} , R_2^{Lie} , and $D_{\text{Lie}}^{(\leq d_{\text{max}})}$;
9. the registry is closed or complete.

The stable formulation is:

Strict admission is an inherited construction. Capability-sound compiler emission is the theorem-level result. Registry census and concrete realizations are certificates or observations at their declared levels. Paper IX owns the ordered-calibration theorem and response certificate; this paper preserves their typed evidence status. Mechanism-based cross-species causality, the Lie/Hall rate hierarchy, and observable sufficiency remain external or downstream Research Programs rather than compiler claims.

The interface therefore supports cross-species reporting only through carrier-qualified typed fields. Depending on declared capabilities, these may include operator, route, word, independently registered Lie/Hall, trajectory, wall, rate, repair, and plateau records.

Capability-Gated Compiler Output

The typed compiler emits only modules supported by each admitted source.

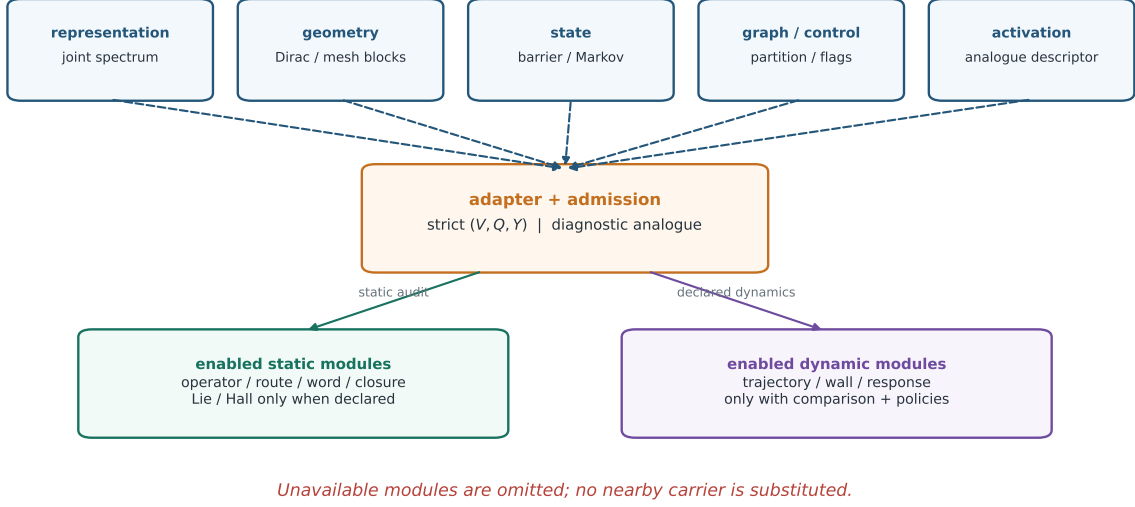


Figure 6. Capability-gated compiler output. Strict sectorization may come from representation theory, geometry, stochastic barriers, or control flags; diagnostic analogues enter through a separate mapping. The compiler output includes only modules supported by the admitted input’s declared capabilities and evidence.

7 Conclusion

This paper establishes capability-aware compilation and Registry evidence. Strict admission factors through the static object language of Paper VIII, while deformation fields and response results are inherited from Paper IX. Under the versioned Manifest, IR, Profile, rule registry, and emission algorithm, capability-sound compilation emits only claims whose carrier, convention, policy, evidence, and derivation checks pass. The Registry demonstrates this interface with sparse strict and analogue records; it is evidence architecture, not the soundness theorem itself.

The result is a common compiler interface rather than a common mathematical dynamics. Its output is not a completed SOFRS report. Paper XI owns wall-record classification, Paper XII the versioned single-report protocol [10], Paper XIII aligned sparse comparison, and Paper XIV policy-relative decision semantics.

8 Outlook

The next compiler questions are:

1. define versioned domain and Registry-to-IR adapters together with scientifically testable adequacy obligations;
2. identify carrier-specific proxy-to-shadow rules with explicit margin, stability, and promotion hypotheses;
3. extend compilation across nontrivial fibre bundles and schema transitions without treating incomparable fields as missing values;

4. enlarge the derivation-rule registry only when theorem hypotheses, certificate requirements, and negative boundaries can be checked mechanically.

Papers XI–XIV consume the resulting typed records through wall, report, alignment, and decision contracts. Those downstream layers do not revise the compiler theorem or supply missing upstream evidence.

Appendix A Computational Artifacts

The following repository artifacts implement the compiler contracts, Registry migration, and source-addressed evidence checks used above.

A.1 Compiler Contracts

The default directory is `schemas/sofcompiler/`. All short paths below are relative to that directory.

Artifact	Role	Short path
A1	Capability Manifest schema	<code>capability-manifest-v1.0.schema.json</code>
A2	Typed SOF IR schema	<code>typed-sof-ir-v1.0.schema.json</code>
A3	Report Profile schema	<code>report-profile-v1.0.schema.json</code>
A4	versioned derivation-rule registry	<code>rule-registry-v1.0.json</code>
A5	compiler examples, typed output regression, and validator	<code>examples/</code> ; <code>examples/strict-associative-compiler-output-v1.0.json</code> ; <code>validate_examples.py</code>

A.2 Registry and Evidence

The short paths in this table are relative to the base directory shown in the second column.

Artifact	Base directory	Role	Short path
A6	<code>registry/</code>	immutable Registry v1.0 snapshot	<code>paper10-release-v1.0.registry.json</code>
A7	<code>registry/</code>	typed Registry v2.0 snapshot	<code>paper10-typed-v2.0.registry.json</code>
A8	<code>schemas/registry/</code> and <code>registry/</code>	Registry schema and snapshot validator	<code>v2.0.schema.json</code> ; <code>validate_snapshot.py</code>
A9a	<code>experiments/paper10/</code>	generated cross-species evidence record	<code>results/registry_evidence_v2.json</code>
A9b	<code>experiments/paper10/</code>	legacy certificate-import record	<code>results/legacy_certificate_imports_v2.json</code>
A10a	<code>experiments/paper10/</code>	cross-species evidence producer	<code>validation/build_results.py</code>
A10b	<code>experiments/paper10/</code>	legacy-import producer	<code>validation/build_legacy_certificate_imports.py</code>
A10c	<code>experiments/paper10/</code>	result validator	<code>validation/validate_results.py</code>
A11	<code>registry/</code> and <code>tests/</code>	v1-to-v2 migrator and migration regression	<code>migrate_v1_to_v2.py</code> ; <code>test_registry_migration.py</code>

A5 checks the contract examples, including claim-local capability, convention, policy, finding-dependency, derivation, and typed emission gates. A8 validates both Registry snapshots by default. A10a–A10c build the two evidence records and validate the main cross-species record. A11 regenerates the v2.0 snapshot from the immutable v1.0 source and checks the migration and legacy-import invariants. The snapshots store artifact identifiers and digests so that findings, certificates, and producer paths remain source-addressed. These executable checks certify the declared implementations and finite records; they do not replace the compiler theorem proof. All listed artifacts are available in the [RIME repository](#).

Artifact A9b is explicitly a migration certificate for two finite values retained from the immutable v1 snapshot; it does not represent a fresh scientific recomputation. The Rubik joint-spectrum and direct-support certificates instead cite their existing Paper IV and Paper II result JSON directly.

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